

Snippet compile check

September 14, 2026

1 Applied classification

Abstract

Someone publishes a number in years — an aquifer with a decade or two left, a phosphate reserve of three centuries — and a reader hears a date. Three different quantities travel under that single label: a ratio of an economic reserve classification to current production, a trend fitted to a satellite gravity anomaly, and a log biomass margin divided by a fishing mortality. Each answers a real question; none is a time at which anything physically runs out. A weighted aggregate compounds the confusion by hiding a component deficit behind a positive scalar.

We build the accounting layer that keeps the three apart. Compartments carry a material identity, a boundary, and a unit, so biomass, money, and biodiversity are never summed into one scalar; conservation follows from the incidence structure of the flux network, positivity from donor limitation, and services are readouts rather than conserved mass. We separate three predicates the literature conflates — accounting consistency, stoichiometric conservation, and barrier admissibility — and prove their relations, so that each claim about a material system carries the predicate it actually establishes. Depletion time splits into gross turnover intensity, a frozen-rate ratio, and a scenario-conditioned hitting time, with uniform-drift bounds between them. No nonnegative weighting of component balances certifies that every component clears its floor: certification needs the vector. We classify three public applications at their exact status and state the surrogate on which each first-passage result is conditioned. We name the gaps that remain open, and we read weak and strong sustainability as two regimes of one system, distinguished by whether the material cycle closes at the rate of use.

Keywords: material flow accounting; stock–flow ledger; depletion indicators; first-passage time; conservation laws; composite indicators; reserve life

Who this paper is for, and in what order. The verdicts, in thirty seconds: the title, this note, and §1. The argument, in three minutes: §1, §6, §10.1, §11. The machinery, in an hour: §2 (the ledger), §3 (three certificates), §4 (what the closed ledger guarantees), §5 (readouts and the componentwise deficit), §6–§7 (depletion time and its passage semantics), §8 (domain templates at registered status), §9 (the interface with institutional delay dynamics). Readers who want only the applied conclusion can take §6.5 and the classification table; the theorems are what license those readings, and nothing else in this paper is required to accept them.

What this paper is not. It is not a forecast, not an index, and not a recommendation about any particular aquifer, fishery, or mine. It is an accounting layer whose purpose is to make sure that when a statement about depletion is made, the statement that is true is the statement that is written down.

1.1 Two ways the bookkeeping goes wrong

A water board announces that a district has a decade or two of groundwater at current rates. A geological survey lists phosphate reserves of roughly 1,000,000 kt, and someone divides by production and announces three centuries. A fisheries assessment divides a log biomass margin by a fishing mortality and calls the quotient a time scale. All three numbers are computed correctly. All three answer different questions, and only the last is even described in the language of its own construction.

Sustainability accounting fails in two characteristic ways. The first is *compensatory aggregation*: heterogeneous physical stocks and service flows are summarised by scalar indices whose cross-component trades are never declared as mathematics, so a severe deficit in one component can coexist with a positive aggregate. The composite-indicator and weak-versus-strong-sustainability literatures document the failure and its noncompensatory remedies (Munda and Nardo, 2009; Ekins et al., 2003; Neumayer, 2013). The second is *classification drift*: quantities carrying the units of time circulate as if they were one quantity, “time to depletion”, though they answer different questions under different assumptions.

We do not dispute the arithmetic of any of these indicators. We dispute the shared label. The two theses are not equivalent, and only the second is defensible for a ledger of this kind.

Four limits, registered rather than conceded later. (i) The groundwater two-pool model is an open gap; the applied object we admit is its one-pool affine approximation (§8.2). (ii) Spawning biomass is not an abiotic support pool, and we do not treat it as one (§6.5.4). (iii) Every first-passage result is conditioned on a declared surrogate and carries explicit non-claims (§7.7). (iv) The numerical exhibits are worked instances of the constructions, not constitutive claims for the named domains. A reader looking for guidance about a specific resource will not find it here; a reader looking to know which of their claims their data can carry will.

We fix the objects first: what a compartment is, what a primitive flux is, and

why a conversion may appear only as an explicit coefficient. Everything that follows is a property of this bookkeeping and nothing else.

Three predicates get conflated in the literature: the ledger balances, the chemistry closes, the barriers hold. We prove each separately so that a reader can tell which one a given argument uses.

Here the ledger pays for itself. Mass is conserved because of the incidence pattern, nothing goes negative because every outflow is donor-limited, and no positive-effort interior rest exists to hide in.

Services are read off the ledger, not added to it. The deficit is a vector, and §5.4 shows exactly where a scalar would have hidden it.

Three quantities, three questions. We keep them named separately through §6.5, where each public indicator is classified at the status its construction supports.

A hitting time is a property of a model, so we state the model, the barrier, and the conditioning, and then say what does not follow (§7.7).

Templates for phosphorus, groundwater, and harvest-side economics, each with its status and its admitted object marked, so a reader can see what is a construction and what is a claim about the world.

One shared object, two systems, and a non-reduction result that says why the connection is an interface contract rather than a limit.

Not caveats: prohibitions on a class of argument. Five double-counting rules, a theorem against compensatory weighting, and a discipline that keeps negative and boundary content first-class.

Nothing in this paper argues that the numbers are wrong. It argues that a reader should be able to tell, from the statement itself, which question was answered. That is a bookkeeping property, and it is cheap to install.